

并查集

人员

李瑞涵、田心一、柳力玮、赵书梵、刘祺、明宗亮、苑钊、纪博涵、蒋叔璋 到课, 杨咏丞、刘宸熙、高健桓 线上

上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4676>

王向东老师周日三点半C++综合练习								
刷新								
#	用户名	姓名	编程分	时间	A (12 / 22)	B (12 / 22)	C (7 / 13)	D (2 / 2)
1	tianxinyi	田心一	4	10:46:24	0:43:11	0:57:36	(-3) 1:53:2	6:12:35 ♥
2	liuliwei	柳力玮	4	56:17:49	0:35:44	0:59:5	(-3) 1:51:40 ♥	51:51:20
3	zhaoshufan	赵书梵	3	3:36:7	(-1) 0:28:30 ♥	0:55:54 ♥	1:51:43	
4	mingzongliang	明宗亮	3	3:53:59	0:32:6	(-1) 0:59:0	2:2:53	
5	yuanzhao	苑钊	3	4:27:0	(-1) 0:53:46	1:8:54	2:4:20	
6	liuqi	刘祺	3	7:28:52	0:44:55	1:23:50	5:20:7	
7	yangyongcheng	杨咏丞	3	128:41:37	(-2) 0:44:55	(-1) 1:35:43	125:20:59	
8	chujinyang	初锦阳	2	2:57:11	0:55:16	(-1) 1:41:55		
9	gaojianhuan	高健桓	2	2:57:39	0:42:25	(-2) 1:35:14		
10	liuchenxi	刘宸熙	2	3:21:35	(-1)	(-2)		

本周作业

<https://cppoj.kids123code.com/contest/4813> (课上讲了 A ~ E 题, 课后作业是 A ~ E 题必做, F G 题选做)

课堂表现

今天给同学们讲了 并查集 这个算法, 并查集 代码非常简洁, 两行模板代码要求同学们必须要写熟。

课堂内容

素数个数

埃氏筛 模板, 当 f[i]==true 时, 说明 i 是合数

原理: 所有质数的倍数都是合数

```
#include <bits/stdc++.h>

using namespace std;
```

```
const int maxn = 1e8 + 5;
bool f[maxn];

int main()
{
    int n; cin >> n;
    for (int i = 2; i <= n; ++i) {
        if (f[i]) continue;
        for (int j = i+i; j <= n; j += i) f[j] = true;
    }

    int res = 0;
    for (int i = 2; i <= n; ++i) {
        if (!f[i]) ++res;
    }
    cout << res << endl;
    return 0;
}
```

【模板】并查集

并查集 模板题

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
int f[maxn];

int fFind(int x) {
    if (f[x] != x) f[x] = fFind(f[x]);
    return f[x];
}

int main()
{
    int n, m; cin >> n >> m;
    for (int i = 1; i <= n; ++i) f[i] = i;

    while (m -- ) {
        int op, x, y; cin >> op >> x >> y;
        if (op == 1) {
            int fx = fFind(x), fy = fFind(y);
            if (fx != fy) f[fx] = fy;
        } else {
            if (fFind(x) == fFind(y)) cout << "Y" << endl;
            else cout << "N" << endl;
        }
    }
}
```

```
    return 0;
}
```

亲戚

并查集 模板题

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 5000 + 5;
int f[maxn];

int fFind(int x) {
    if (f[x] != x) f[x] = fFind(f[x]);
    return f[x];
}

int main()
{
    int n, m, p; cin >> n >> m >> p;
    for (int i = 1; i <= n; ++i) f[i] = i;

    while (m -- ) {
        int x, y; cin >> x >> y;
        int fx = fFind(x), fy = fFind(y);
        if (fx != fy) f[fx] = fy;
    }

    while (p -- ) {
        int x, y; cin >> x >> y;
        if (fFind(x) == fFind(y)) cout << "Yes" << endl;
        else cout << "No" << endl;
    }
    return 0;
}
```

村村通

一共 n 个城市, 全合并起来需要 $n-1$ 次

可以设 $\text{cnt} = n-1$, 之后每合并一次就让 $\text{cnt}--$ 就可以了

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1000 + 5;
```

```

int f[maxn];

int fFind(int x) {
    if (f[x] != x) f[x] = fFind(f[x]);
    return f[x];
}

int n, m;

void solve() {
    for (int i = 1; i <= n; ++i) f[i] = i;
    int cnt = n - 1;
    while (m -- ) {
        int x, y; cin >> x >> y;
        int fx = fFind(x), fy = fFind(y);
        if (fx != fy) { f[fx] = fy; --cnt; }
    }
    cout << cnt << endl;
}

int main()
{
    while (true) {
        cin >> n;
        if (n == 0) break;
        cin >> m;
        solve();
    }
    return 0;
}

```

集合

先用 埃氏筛 找出所有的质数, 然后枚举比 p 大的质数 i , 找到 $a \sim b$ 中所有 p 的倍数, 将他们利用并查集合并。

```

#include <bits/stdc++.h>

using namespace std;

const int maxn = 1e5 + 5;
int f[maxn];
bool st[maxn];

int fFind(int x) {
    if (f[x] != x) f[x] = fFind(f[x]);
    return f[x];
}

int main()
{
    for (int i = 2; i < maxn; ++i) {

```

```

    if (!st[i]) {
        for (int j = i+i; j < maxn; j += i) st[j] = true;
    }
}

int a, b, p; cin >> a >> b >> p;
for (int i = a; i <= b; ++i) f[i] = i;

int cnt = b - a + 1;
for (int i = p; i < maxn; ++i) {
    if (st[i]) continue;

    vector<int> vec;
    for (int j = i; j < maxn; j += i) {
        if (j>=a && j<=b) vec.push_back(j);
    }

    for (int j = 1; j < (int)vec.size(); ++j) {
        int fx = fFind(vec[0]), fy = fFind(vec[j]);
        if (fx != fy) f[fx] = fy, --cnt;
    }
}
cout << cnt << endl;
return 0;
}

```

[SCOI2005] 繁忙的都市

二分 + 并查集 判断

```

#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
int f[maxn];
struct node {
    int u, v, value;
    bool operator < (const node& p) const { return value < p.value; }
};
vector<node> vec;
int n, m;

int fFind(int x) {
    if (f[x] != x) f[x] = fFind(f[x]);
    return f[x];
}

bool check(int mid) {
    for (int i = 1; i <= n; ++i) f[i] = i;

```

```
int cnt = n;
for (node it : vec) {
    int u = it.u, v = it.v, value = it.value;
    if (value > mid) continue;

    int pu = fFind(u), pv = fFind(v);
    if (pu != pv) f[pu] = pv, --cnt;
}

return cnt==1;
}

int main()
{
    cin >> n >> m;
    while (m -- ) {
        int u, v, value; cin >> u >> v >> value;
        vec.push_back({u, v, value});
    }

    int l = 1, r = 10000;
    while (l <= r) {
        int mid = (l + r) / 2;
        if (check(mid)) r = mid-1;
        else l = mid+1;
    }
    cout << n-1 << " " << l << endl;
    return 0;
}
```